Web Server Deployment

SSH service (Key based Authentication)

1. SSH (Secure Shell) is a network communication protocol used to securely connect two computers over an unsecured network.
2. It uses encryption (cryptography) to protect data during communication.
3. Commonly used for remote login to servers (like AWS EC2).

Two ways to get ssh

1. **password-based authentication**
   - In password-based authentication we have to assign a password to authenticate
2. **key based authentication**
   - While in key-based authentication we need keys to connect

Key-based authentication

In key based authentication there are two keys

1. Private key

1. Private key basically in encrypted format and these key use by the user to authenticate with the server
2. This file having the extension { .pem }
3. Do not share the private key

2.Public key

1. While the public key stored by the server to match with the private key of the user
2. We can share this key openly

What is webserver

A web server is software (and sometimes hardware) that stores, processes, and delivers web pages to users over the internet.

Popular webserver

1.apache
2.nginx
3.webserver IIS

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HTTP (Hypertext Transfer Protocol):

1. It is a protocol used to transfer data (web pages) between a browser and a web server.
2. It works on the port 80

HTTPs (Hypertext Transfer Protocol Secure):

1. It is the secure version of HTTP, where data is encrypted.
2. While it works on the port 443

HTTP/HTTPS is a communication protocol between the browser and the web server.

Deploy web server using NGINX

1.create a instance

1.name ----------> webserver
2.machine ----------> ubuntu
3.key name ---------> ubuntu
4.instance type ----> t3. Micro

2.get ssh of the instance

1. ssh -I <key_name> <machine user>@<public_ip>
2. apt update -y
3. apt install nginx -y
4. systemctl start nginx
5. cd /var/www/html/
6. rm -rf index.html
7. vim index.html
   </h1> My 1st Deployment </h1>
8. systemctl restart nginx

3.<public_ip> ----------> hit on the web

Deploy CSS Template on Apache

1.create a instance

1.name ----------> webserver
2.machine ----------> ubuntu
3.key name ---------> ubuntu

4.instance type ----> t3. Micro

2.get ssh of the instance

1. ssh -I <key_name> <machine user>@<public_ip>
2. apt update -y
3. apt install apache2 -y
4. systemctl start apache2
5. wget --content-disposition https://templatemo.com/download/templatemo_610_aurum_gold
6. apt install zip -y
7. unzip templatemo_610_aurum_gold.zip
8. rm -rf templatemo_610_aurum_gold.zip
9. cp -rf templatemo_610_aurum_gold/* /var/www/html/
10. systemctl restart apache2
11. systemctl enable apache2

3.<public_ip> ----------> hit on the web

Create a Instance Windows & install webserver IIS

1. create a instance

1.name----------> Windows_Server-2016-English-Full-Base-2026.02.11
2.instance type------> t3. micro
3.security group----> ssh / http / RDP (3389)
4.create a instance

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2.access the instance

1.connect the instance
2.go to RDP client
3.click on get password------> browse key pair-----> decrypt password ---> copy
4.open RDP in desktop
5.computer---> (public IP)
6.username---> (Administrator)
7.password----> (paste password)
8.connect

3.Install webserver IIS and create a webpage on windows server

1.search server manager and click
2.add roles and features
3.next

4.role based or features based installation
5.next
6.select a server from server pool
7.next
8.select one file------> webserver IIS------add features-------next------next-----next-----install-----> close
9.click on file manager
10.click on c drive
11.inetpub-----> wwwroot ------> 2files (delete these files)
12.create a index.html file but open the notepad ---> save as index.html ---> all files "HELLO WORLD"

4.(public IP) ---------->check the web

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